Article

Cryogenic Feedstock as a Cold-Trap Heat Sink: Condenser-to-Feed Heat Integration for Lunar Polar Water Extraction

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Abstract: Thermal extraction of water from lunar polar regolith requires heating large masses of cryogenic soil and then removing the same latent heat again at a cold trap. Published system designs reject that condensation heat to radiators or chillers. This article proposes routing it instead into the incoming feedstock through an indirect heat exchanger. Because regolith excavated from permanently shadowed regions arrives near 40 K, well below a typical cold-trap temperature of 200–220 K, the transfer is spontaneous and needs no heat pump. A steady-state heat balance using the temperature-dependent heat capacity of Apollo soils shows that condenser-to-feed integration alone reduces the ideal extraction heat by 17–38% across ice contents of 1–10% by mass. Combined with conventional recuperation from spent regolith, the reduction is 21–59%. Integration also compresses the energy penalty of low-grade deposits: ideal heat varies by a factor of 1.6 between 1% and 10% ice, against 3.0 without heat recovery. Below about 1.4–2.4% ice, depending on cold-trap temperature, the feed can absorb all condensation heat and the cold-trap radiator can be eliminated. A targeted review of the literature found no prior description of the concept. A bench experiment to measure the governing heat-transfer coefficient is proposed.

Keywords: in-situ resource utilization; lunar water ice; permanently shadowed regions; thermal extraction; cold trap; heat integration; regolith

1. Introduction

Water ice in the permanently shadowed regions of the lunar poles is the leading candidate feedstock for propellant, life support and radiation shielding on the Moon. Its value depends heavily on the energy required to recover it. Most proposed recovery methods are thermal: icy regolith is heated, the ice sublimes, and the vapour is collected by deposition on a cold surface. Every such process carries three thermal loads:

1. **Sensible heat** to warm the soil and ice from about 40 K to the extraction temperature.
2. **Latent heat** of sublimation, about 2.84 MJ per kilogram of water.
3. **Rejected heat** at the cold trap, where the same latent heat is released as the vapour redeposits.

Because polar regolith is expected to hold only a few percent ice by mass, the first load scales with roughly 20–100 kg of soil per kilogram of water and can exceed the latent heat. The third load is an engineering burden of its own: at a production rate of 15,000 tonnes of water a year, the cold trap must reject about 1.35 MW, requiring on the order of a hectare of ideal radiator.

Heat recovery from hot spent regolith has been studied for oxygen-extraction reactors, notably the counterflow regolith heat exchanger developed for NASA [1]. The cold trap's heat, by contrast, has been treated as waste in the water-extraction designs reviewed here [2–5]. This article examines the opposite treatment: using the cryogenic feedstock as the cold trap's heat sink.

2. Concept

The proposed process passes the feedstock through three heat-exchange stages in order of rising temperature, so that each heat source is used where it is warmer than the stream it heats (Figure 1).

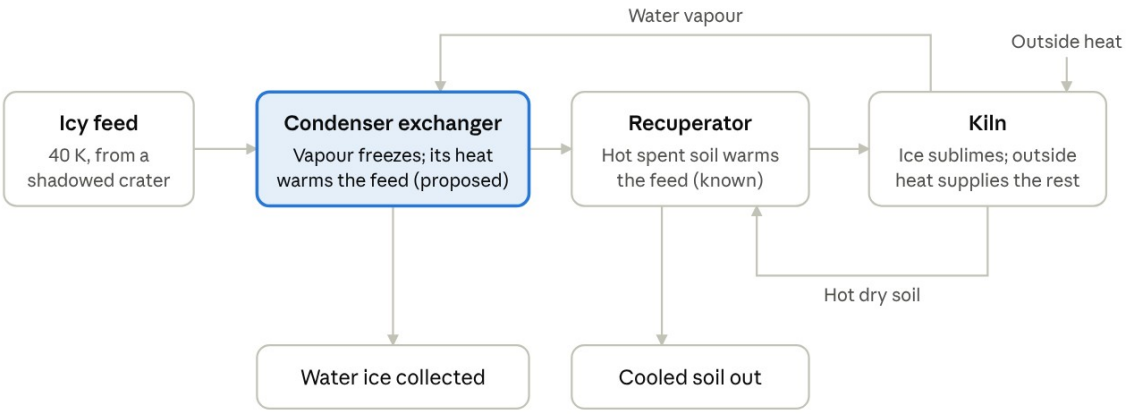


Figure 1. Heat-integrated extraction process. Deposition heat released in the condenser exchanger (highlighted) warms the incoming 40 K feed; a spent-soil recuperator adds further heat before the kiln supplies the remainder.

In the first stage, the cold trap is built as an indirect heat exchanger: vapour deposits on one side of a wall while the 40 K feed moves past the other side. The deposition heat crosses the wall and raises the feed toward the cold-trap temperature less a temperature approach. In the second stage, a counterflow recuperator transfers heat from the hot, dried soil leaving the kiln. The kiln then supplies only the remaining sensible heat and the latent heat of sublimation.

Two design constraints follow from the physics:

- **Vapour and feed must be separated by a wall.** In direct contact, the cold feed would itself act as the cold trap; deposited ice would return to the kiln and circulate without being collected.
- **The preheater must be sealed or held below about 170 K.** Above that temperature, ice in the feed begins to sublime at appreciable rates. In a sealed system the vapour still reaches the cold trap; in an open hopper it is lost.

The second law bounds the benefit. Heat released at 200–220 K can raise the feed only to about 180–200 K and cannot supply the kiln at 250 K or above without a heat pump. The latent heat of sublimation, 0.79 kWh per kilogram of water, is therefore a floor that no arrangement of passive heat exchange can go below.

3. Model and Assumptions

The analysis is a steady-state energy balance per kilogram of water collected, with no wall losses, so all results are ideal lower bounds. For an ice mass fraction w, each kilogram of water is accompanied by $m_r = (1 - w)/w$ kilograms of dry regolith. The external heat supplied at the kiln is

$$Q_{\text{ext}} = m_r \int_{T_0}^{T_x} c_{p,\,\text{reg}}\, dT + \int_{T_0}^{T_x} c_{p,\,\text{ice}}\, dT + L_{\text{sub}} - Q_{\text{cond}} - Q_{\text{rec}}$$

where the heat recovered from the condenser is limited by what the feed can absorb below the cold-trap temperature:

$$Q_{\text{cond}} = \min\left(L_{\text{sub}},\; m_r \int_{T_0}^{T_c - \Delta T} c_{p,\,\text{reg}}\, dT + \int_{T_0}^{T_c - \Delta T} c_{p,\,\text{ice}}\, dT\right)$$

The recuperator duty Q_{rec} equals its effectiveness ε multiplied by the smaller of the heat the spent soil can release and the heat the preheated feed can still absorb. Condensation heat not absorbed by the feed, $L_{\text{sub}} - Q_{\text{cond}}$, must be rejected by radiator. Model inputs are listed in Table 1.

Table 1. Model inputs.

Parameter	Base value	Basis
Feed temperature T_0	40 K	Typical floor temperature of permanently shadowed regions
Extraction temperature T_x	250 K (varied to 473 K)	Rapid sublimation in vacuum
Cold-trap temperature T_c	220 K (varied 180–240 K)	Assumed; within the range of published cold-trap designs
Exchanger temperature approach ΔT	20 K	Assumed
Recuperator effectiveness ε	0.7	Assumed for a counterflow solids exchanger
Latent heat of sublimation L_{sub}	2.84 MJ/kg	Standard value
Regolith heat capacity	Polynomial fit to Apollo soils [9]	Within about 6% down to 90 K; extrapolated below
Ice heat capacity	Linear fit to measured data [10]	Approximate below 100 K

The strong temperature dependence of regolith heat capacity is central to the result. It rises from about 80 J kg^{-1} K^{-1} at 40 K to 670 J kg^{-1} K^{-1} at 250 K, so warming dry regolith from 40 K to 250 K requires only 85 kJ/kg, and much of that warming lies below typical cold-trap temperatures. The model is implemented in Python (see Data Availability Statement).

3.1. Use of AI Tools

The concept, the numerical model, the literature search and the manuscript text were developed with the assistance of a generative AI tool based on a large language model, under the author's direction.

4. Results

4.1. Extraction Energy

With condenser integration and recuperation combined, ideal external heat lies between 0.84 and 1.30 kWh per kilogram of water across 1–10% ice, against 1.07–3.19 kWh/kg for an unintegrated kiln (Figure 2).

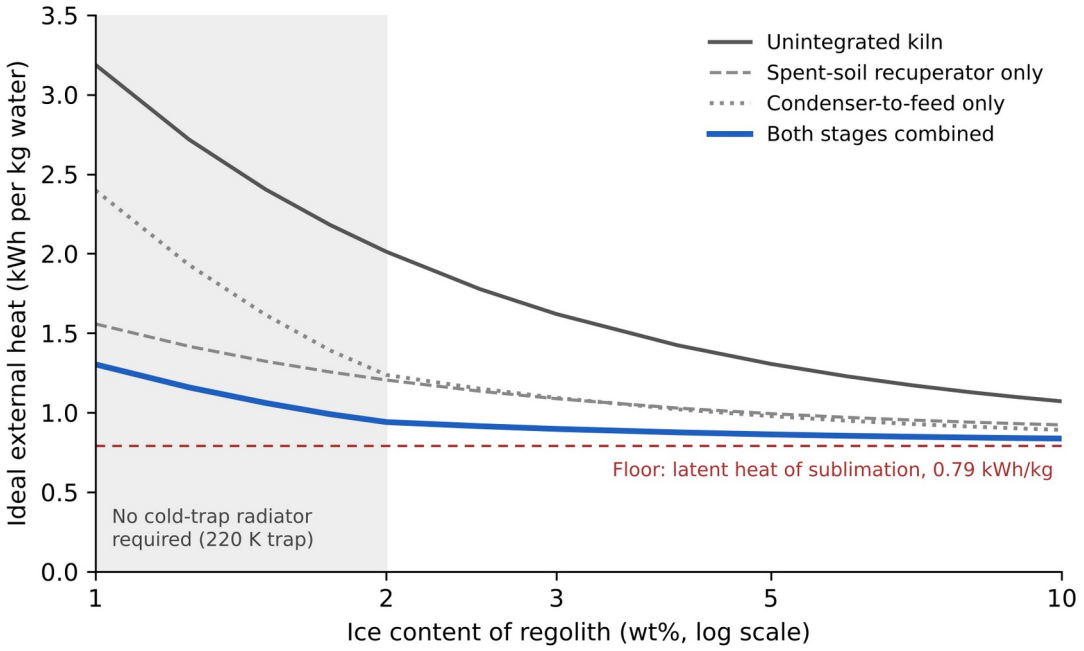


Figure 2. Ideal external heat per kilogram of water collected as a function of ice content, for extraction at 250 K, a 220 K cold trap, a 20 K exchanger approach and recuperator effectiveness 0.7. The shaded band marks ice contents at which the feed absorbs all condensation heat.

Condenser integration alone performs comparably to a spent-soil recuperator alone between 2% and 10% ice, and less well at 1%, where the fixed latent heat is spread over a large soil mass and preheats it only to about 150 K. Applied in addition to a recuperator, it removes a further 9–22% of the external heat. The combined curve approaches the latent-heat floor at high ice content, so further sensible-heat recovery offers diminishing returns there.

4.2. Radiator Elimination

The feed can absorb all condensation heat when its sensible-heat capacity between 40 K and $T_c - \Delta T$ exceeds the latent heat. This occurs below a crossover ice content that depends on cold-trap temperature (Table 2).

Table 2. Crossover ice content below which no cold-trap radiator is required.

Cold-trap temperature	Crossover ice content
190 K (preheat capped at 170 K)	1.4%
200 K	1.6%
220 K	2.0%
240 K	2.4%

Above the crossover, the radiator shrinks in proportion to the heat absorbed. At 5% ice and a 220 K cold trap, the feed absorbs 42% of the condensation heat.

4.3. Sensitivity

Table 3. Ideal external heat at 5% ice under varied conditions.

Condition	Unintegrated (kWh/kg)	Both stages (kWh/kg)	Reduction
Extraction at 250 K (base case)	1.31	0.86	34%
Extraction at 373 K	1.89	1.10	42%
Extraction at 473 K	2.56	1.36	47%
60% of ice recovered	1.65	0.90	45%
Preheat capped at 170 K	1.31	0.90	31%

The benefit grows with extraction temperature and with incomplete recovery, both of which increase the sensible heat carried by the soil. Capping the preheat to avoid premature sublimation costs only three percentage points.

5. Discussion

5.1. Relation to Prior Work

A targeted search of journal articles, NASA technical reports and patents (November 2026) found no prior description of condenser-to-feed heat integration in lunar water extraction. Designs that address cold-trap heat reject it: NASA's Lunar Auger Dryer ISRU sends it to a chiller in ground testing and to a radiator for lunar operation, and chills its feed hopper to keep the soil frozen [2]; the Advanced Cooling Technologies and Honeybee Robotics ice-miner design rejects it by radiation through a variable-conductance heat-pipe cold trap [3]; and an americium-powered mining rover rejects it through radiator fins [4]. A 2025 review of lunar water extraction technologies does not discuss heat recovery but identifies the integration of extraction and vapour-capture hardware as a research gap [5]. Recuperation from spent regolith is established prior art [1]. The search did not cover conference abstract archives, paywalled full texts or non-English literature, and the claim of novelty should be read with that limitation.

An alternative line of work avoids heating the bulk soil altogether by separating ice grains mechanically and electrostatically before any phase change [12,13]. Where such beneficiation is feasible, it reduces the soil mass that must be heated and therefore the benefit of heat integration. The two approaches are complementary: any residual thermal step on a concentrate would still produce condensation heat that could warm the incoming material.

5.2. Comparison with Demonstrated Systems

Published prototypes consume far more energy than the ideal balance requires (Table 4).

Table 4. Measured energy of published prototypes against the model's ideal value.

System	Measured (kWh per kg water)	Ideal value as a share of measured
Drill-based thermal extraction [6]	8.6	11%
Microwave heating, best case at 75% recovery [7]	2.0	48%
LUWEX integrated system, regolith simulant [8]	about 44	2%

The ideal figure is 0.97 kWh/kg for 6% ice starting at −80 °C, the conditions of the drill-based test. The gap indicates that heat losses, incomplete recovery and slow heat transfer currently dominate extraction

energy. Heat integration addresses the thermodynamic share of the budget; its practical value will depend on those losses being reduced in parallel.

5.3. Limitations

- **Heat transfer rate.** The model sets no limit on how fast heat crosses the exchanger wall. Regolith conducts heat very poorly in vacuum, so the exchanger area, residence time and any need for agitation or a low-pressure gas to improve contact are the principal engineering unknowns.
- **Heat capacity below 90 K.** The regolith fit is extrapolated in this range. Semi-empirical models suggest heat capacity there may be higher than standard fits predict [11], which would favour the concept.
- **Ice composition.** Polar ice likely contains sulfur, nitrogen and carbon compounds that deposit at different temperatures and may foul exchanger surfaces.
- **Steady state.** Batch operation, start-up and transient behaviour are not modelled.
- **Radiator sizing.** Area estimates assume an ideal emitter facing cold sky; real radiators near sunlit or warm terrain would be larger, which strengthens the case for eliminating them.

6. Proposed Experiment

The decisive unknown is the heat-transfer coefficient between a cold-trap wall and a moving bed of cryogenic regolith in vacuum. A bench test can measure it directly:

1. Prepare lunar regolith simulant with 2% and 5% water ice by mass and cool it to 77 K with liquid nitrogen in a thermal-vacuum chamber.
2. Sublime vapour from a separately heated icy sample onto a cold plate whose rear face contacts a slowly moving bed of the cold simulant.
3. Measure the heat flux through the plate, the temperature rise of the bed, the mass of ice deposited and the plate temperature, with the bed in contact and with it replaced by an equivalent cryocooler load.
4. Repeat with light agitation of the bed and with a low pressure of non-condensable gas around it, to quantify the improvement each gives.

The outputs, a wall-to-bed heat-transfer coefficient and the measured fraction of deposition heat absorbed by the feed, are sufficient to size an exchanger and to test the predictions in Section 4.

7. Conclusions

The cold trap of a lunar water-extraction plant operates far above the temperature of the soil being mined. Using the incoming soil as the cold trap’s heat sink is thermodynamically sound, needs no active heat pump, and in an ideal analysis reduces extraction heat by up to 38% on its own and up to 59% with conventional recuperation. Its most distinctive effects are at low ice content, where it removes the need for a cold-trap radiator and narrows the energy gap between poor and rich deposits. Whether these gains survive contact with real hardware depends on heat transfer into cryogenic granular material in vacuum, which a straightforward laboratory experiment can resolve.

Supplementary Materials

ice_model.py: Python model that reproduces the model results reported in Sections 3 and 4.

Data Availability Statement

The model results can be reproduced with the supplementary file ice_model.py. No other data were generated.

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Conflicts of Interest

The author declares no conflicts of interest.

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